

Name: Mewada Maitrri Manishkumar

Topic: Linux User Management, Sudo, SSH Security, SELinux, and AppArmor

1. Create a new user called streamer and a group called musicfans on your Linux system, then add streamer to the musicfans group. Verify the group membership using the id command.

⇒ Step 1: Create the group

`sudo groupadd musicfans`

Step 2: Create the user

`sudo useradd -m streamer`

(Optional) Set a password:

`sudo passwd streamer`

Step 3: Add the user to the group

`sudo usermod -aG musicfans streamer`

Step 4: Verify group membership

`id streamer`

2. Give your current user sudo privileges by editing the `/etc/sudoers` file using visudo. Test by running a command that requires sudo, such as updating the package list.

⇒ Open the sudoers file safely:

`sudo visudo`

Add the following line (replace username with your username):

`username ALL=(ALL) ALL`

Save and exit.

Test sudo access

On Ubuntu/Debian:

`sudo apt update`

On RHEL/CentOS:

`sudo dnf check-update`

3. Set up SSH key-based authentication: generate an SSH key pair using ssh-keygen, then copy your public key to a remote Linux server (or a local VM) using ssh-copy-id. Disable password authentication for SSH in /etc/ssh/sshd_config and restart the SSH service.

Hint: Look for the PasswordAuthentication setting in the SSH config file.

⇒ ssh-keygen

Press Enter to accept the default file location.

Example:

Generating public/private rsa key pair.

Step 2: Copy the public key to the remote server

ssh-copy-id username@server_ip

Example:

ssh-copy-id student@192.168.1.100

Step 3: Log in using the SSH key

ssh username@server_ip

Edit the SSH server configuration

sudo vi /etc/ssh/sshd_config

Find:

PasswordAuthentication yes

Change it to:

PasswordAuthentication no

Restart the SSH service

Ubuntu:

sudo systemctl restart ssh

RHEL:

```
sudo systemctl restart sshd
```

Step 6: Verify the service

```
Sudo systemctl status sshd
```

4. Check if SELinux or AppArmor is enabled on your system. If SELinux is active, use the `getenforce` command to display its current mode, and switch it to permissive mode. If AppArmor is active, list all enforced profiles using `sudo apparmor_status`.

⇒ If SELinux is active

Check its status:

```
getenforce
```

Example output:

Enforcing

Switch to permissive mode:

```
sudo setenforce 0
```

Verify:

```
Getenforce
```

If AppArmor is active

Check status:

```
sudo apparmor_status
```

5. Use ChatGPT to explain the difference between SELinux and AppArmor in one paragraph, then write a summary in your own words and mention one real-world app (like Zomato or Paytm) that would benefit from these security modules.

⇒ ChatGPT Explanation

SELinux (Security-Enhanced Linux) and AppArmor are Linux security modules that enforce mandatory access control (MAC) to limit what applications and users can do, reducing the impact of compromised software. SELinux uses detailed policies based on labels assigned to files, processes, and other system objects, making it highly flexible but more

complex to configure. AppArmor uses path-based profiles that specify which files and resources an application can access, making it generally simpler to understand and manage. Both improve system security, but SELinux is commonly the default on Red Hat Enterprise Linux, while AppArmor is commonly used on Ubuntu and SUSE systems.

⇒ My Summary

SELinux and AppArmor both protect Linux systems by restricting application access to files and system resources. SELinux provides more detailed, label-based security controls, while AppArmor uses easier-to-manage path-based profiles. Both help reduce security risks if an application is compromised.